WORKED SOLUTION

STEP II 2003, Question 3

Pure • Irrational Numbers and Induction

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s2-q3

The key idea

Irrationality is a "not" property, so the natural tool is the contrapositive: instead of showing that an irrational number has an irrational cube root, show that a rational cube root forces the number itself to be rational. That single fact then drives the induction, because each u_{n+1} is the cube root of u_n. For the last part, we notice that $u_n \to 1$ and slide the whole sequence along by an integer, which keeps every term irrational.

Solution

Cube roots of irrational numbers

Let x be a real number and let $y = \sqrt[3]{x}$ be its real cube root, so $y^3 = x$. We prove the contrapositive: if y is rational, then x is rational.

Suppose $y = \dfrac{p}{q}$ with p, q integers and $q \neq 0$. Then

$$x = y^3 = \frac{p^3}{q^3},$$

and p^3 and q^3 are integers with $q^3 \neq 0$. So x is a ratio of two integers, that is, rational.

So a rational cube root can only come from a rational number. Equivalently, if x is irrational then $\sqrt[3]{x}$ cannot be rational, so it is irrational. This is what we had to prove.

? **Ask yourself** Why not attack the statement directly?

The direct route would start from "x is not a ratio of integers", and there is nothing concrete to calculate with. The contrapositive starts from "$y = p/q$", which hands us integers to work with, and the proof is then one line. Whenever the hypothesis says that something is *not* rational, it is worth trying the contrapositive or a proof by contradiction.

The induction

Since $3^{n+1} = 3 \cdot 3^n$, the laws of indices give

$$u_{n+1} = 5^{1/3^{n+1}} = \left(5^{1/3^n}\right)^{1/3} = \sqrt[3]{u_n}.$$

So each term is the cube root of the one before.

Base case. $u_1 = 5^{1/3} = \sqrt[3]{5}$, which we are told is irrational.

Inductive step. Suppose u_k is irrational for some positive integer k. Then $u_{k+1} = \sqrt[3]{u_k}$ is the cube root of an irrational number, so by the first part it is irrational.

Hence, by induction, u_n is irrational for every positive integer n.

A sequence converging to m

First we find the limit of u_n. Taking logarithms,

$$\ln u_n = \frac{\ln 5}{3^n} \to 0 \quad \text{as } n \to \infty,$$

and since the exponential function is continuous, $u_n = \mathrm{e}^{\ln u_n} \to \mathrm{e}^0 = 1$. Also $\ln u_n > 0$, so $u_n > 1$ for every n; the terms decrease towards 1:

$$u_1 \approx 1.710, \quad u_2 \approx 1.196, \quad u_3 \approx 1.061, \quad u_4 \approx 1.020, \ \ldots$$

> **Ask yourself** How do we move the limit from 1 to m without spoiling irrationality?

Adding an integer to an irrational number gives an irrational number. Indeed, if a is an integer and $u + a = r$ were rational, then $u = r - a$ would be rational too, since a difference of rationals is rational.

So, for a given integer m, define

$$a_n = u_n + m - 1 = 5^{1/3^n} + m - 1.$$

Since $m - 1$ is an integer and u_n is irrational, each a_n is irrational, and

$$a_n \to 1 + m - 1 = m \quad \text{as } n \to \infty.$$

So (a_n) is an infinite sequence of irrational numbers converging to m. This works for every integer m, including $m = 0$.

(Another natural choice, $m\,u_n$, is also irrational and tends to m, but only when $m \neq 0$: for $m = 0$ every term is 0, which is rational. That is why the shift by $m - 1$ is the safer answer.)

Checks

- Numerically, $u_2^3 \approx 1.19581^3 \approx 1.70998 \approx u_1$, confirming $u_{n+1} = \sqrt[3]{u_n}$.

- With $m = 4$: $a_1 \approx 4.710$, $a_2 \approx 4.196$, $a_3 \approx 4.061$, and a_{20} agrees with 4 to more than eight decimal places.
- The difference $a_n - m = u_n - 1$ is positive and roughly $\ln 5/3^n$, so it shrinks by a factor of about 3 at each step.

Common slips

- Proving the converse ("the cube of an irrational is irrational"), which is false: $\left(\sqrt[3]{2}\right)^3 = 2$.
- Using $m\,u_n$ without noticing that it fails for $m = 0$.
- Forgetting to justify that $u_n \to 1$; "the power tends to zero" needs a word about continuity, or the logarithm argument above.

Beyond STEP

This question shows that the irrational numbers are *dense* in the real line: every real number, not just every integer, is a limit of irrationals (try $q_n + \sqrt{2}/n$ with rationals $q_n \to x$). The rationals are dense too, yet in a precise sense there are far more irrationals than rationals: the rationals are countable and the reals are not. Abbott's *Understanding Analysis* covers density, countability and limits of sequences beautifully in its opening chapters.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.